RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824–#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

RA8P1 IO allocation

File: mcu_interfaces.kicad_sch

RA8P1 internal core regulator

File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies

File: mcu_supply_power.kicad_sch

RA8P1 clocks, reset and debug

File: mcu_clocks_debug.kicad_sch

Sheet: /
File: ereader_rev1.kicad_sch

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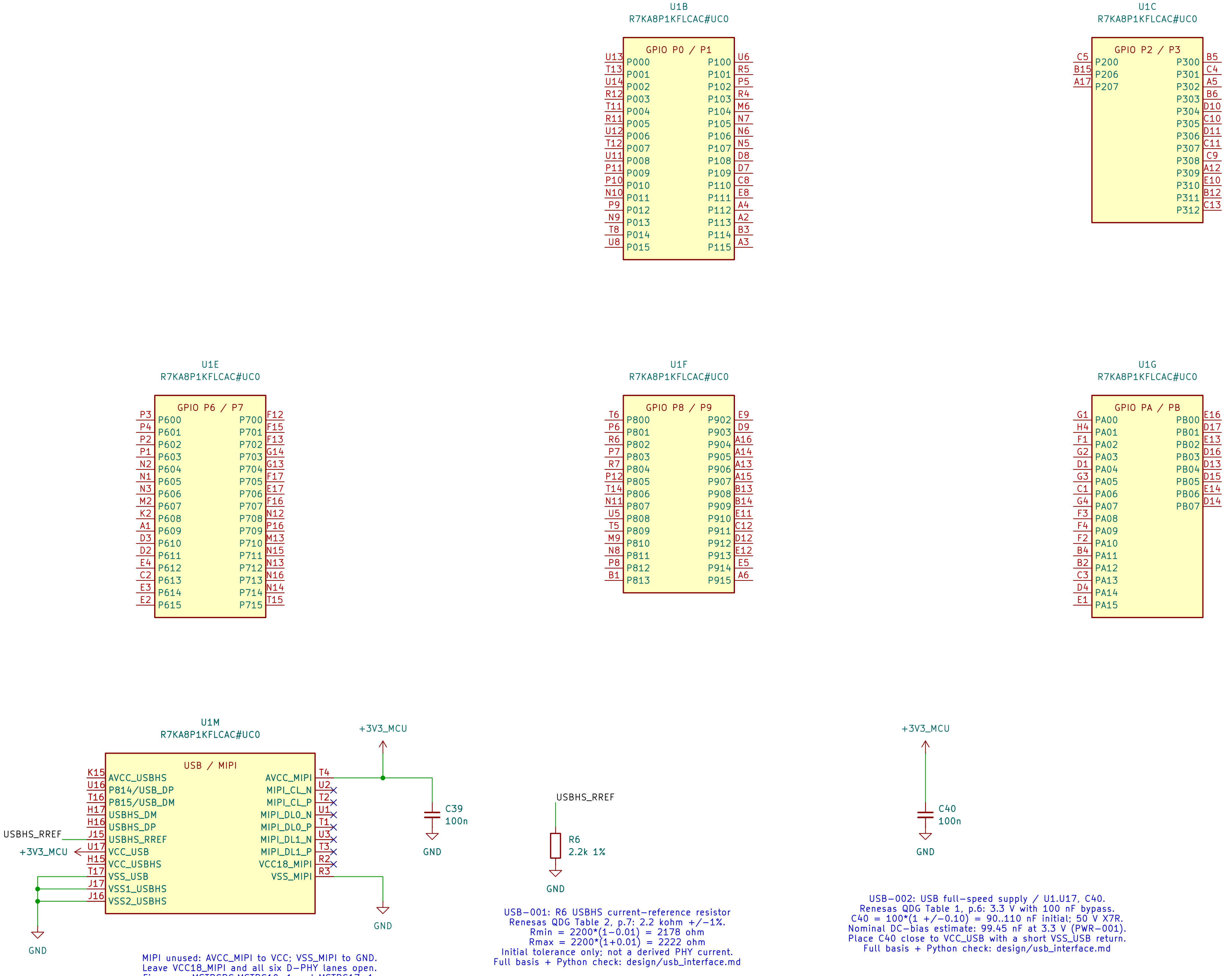
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.

USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md

USB-002: USB full-speed supply / U1.U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90..110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90..110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

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Sheet: /RA8P1 IO allocation/
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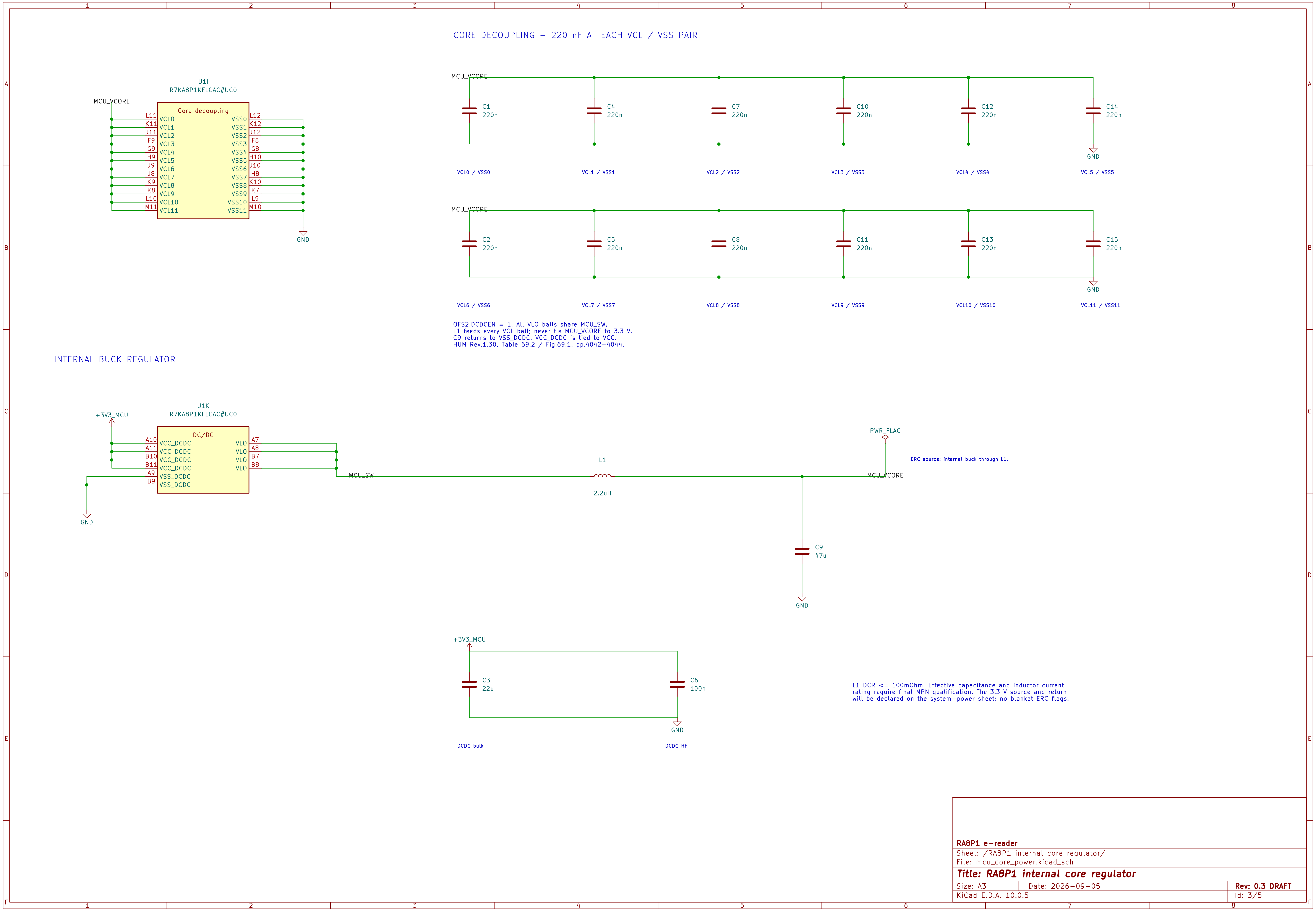

CORE DECOUPLING – 220 nF AT EACH VCL / VSS PAIR
U1I
R7KA8P1KFLCAC#UC0
MCU_VCORE
Core decoupling
L11 VCL0
K11 VCL1
J11 VCL2
F9 VCL3
G9 VCL4
H9 VCL5
J9 VCL6
J8 VCL7
K9 VCL8
K8 VCL9
L10 VCL10
M11 VCL11
VSS0 L12
VSS1 K12
VSS2 J12
VSS3 F8
VSS4 G8
VSS5 H10
VSS6 J10
VSS7 H8
VSS8 K10
VSS9 K7
VSS10 L9
VSS11 M10
GND
MCU_VCORE
C1 220n
C4 220n
C7 220n
C10 220n
C12 220n
C14 220n
GND
VCL0 / VSS0
VCL1 / VSS1
VCL2 / VSS2
VCL3 / VSS3
VCL4 / VSS4
VCL5 / VSS5
MCU_VCORE
C2 220n
C5 220n
C8 220n
C11 220n
C13 220n
C15 220n
GND
VCL6 / VSS6
VCL7 / VSS7
VCL8 / VSS8
VCL9 / VSS9
VCL10 / VSS10
VCL11 / VSS11
OFS2.DCDCEN = 1. All VLO balls share MCU_SW.
L1 feeds every VCL ball; never tie MCU_VCORE to 3.3 V.
C9 returns to VSS_DCDC. VCC_DCDC is tied to VCC.
HUM Rev.1.30, Table 69.2 / Fig.69.1, pp.4042–4044.
INTERNAL BUCK REGULATOR
U1K
R7KA8P1KFLCAC#UC0
+3V3_MCU
DC/DC
A10 VCC_DCDC
A11 VCC_DCDC
B10 VCC_DCDC
B11 VCC_DCDC
A9 VSS_DCDC
B9 VSS_DCDC
VLO A7
VLO A8
VLO B7
VLO B8
GND
MCU_SW
L1
2.2uH
PWR_FLAG
ERC source: internal buck through L1.
MCU_VCORE
C9 47u
GND
+3V3_MCU
C3 22u
C6 100n
GND
DCDC bulk
DCDC HF
L1 DCR <= 100mOhm. Effective capacitance and inductor current
rating require final MPN qualification. The 3.3 V source and return
will be declared on the system–power sheet; no blanket ERC flags.
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Sheet: /RA8P1 internal core regulator/
File: mcu_core_power.kicad_sch
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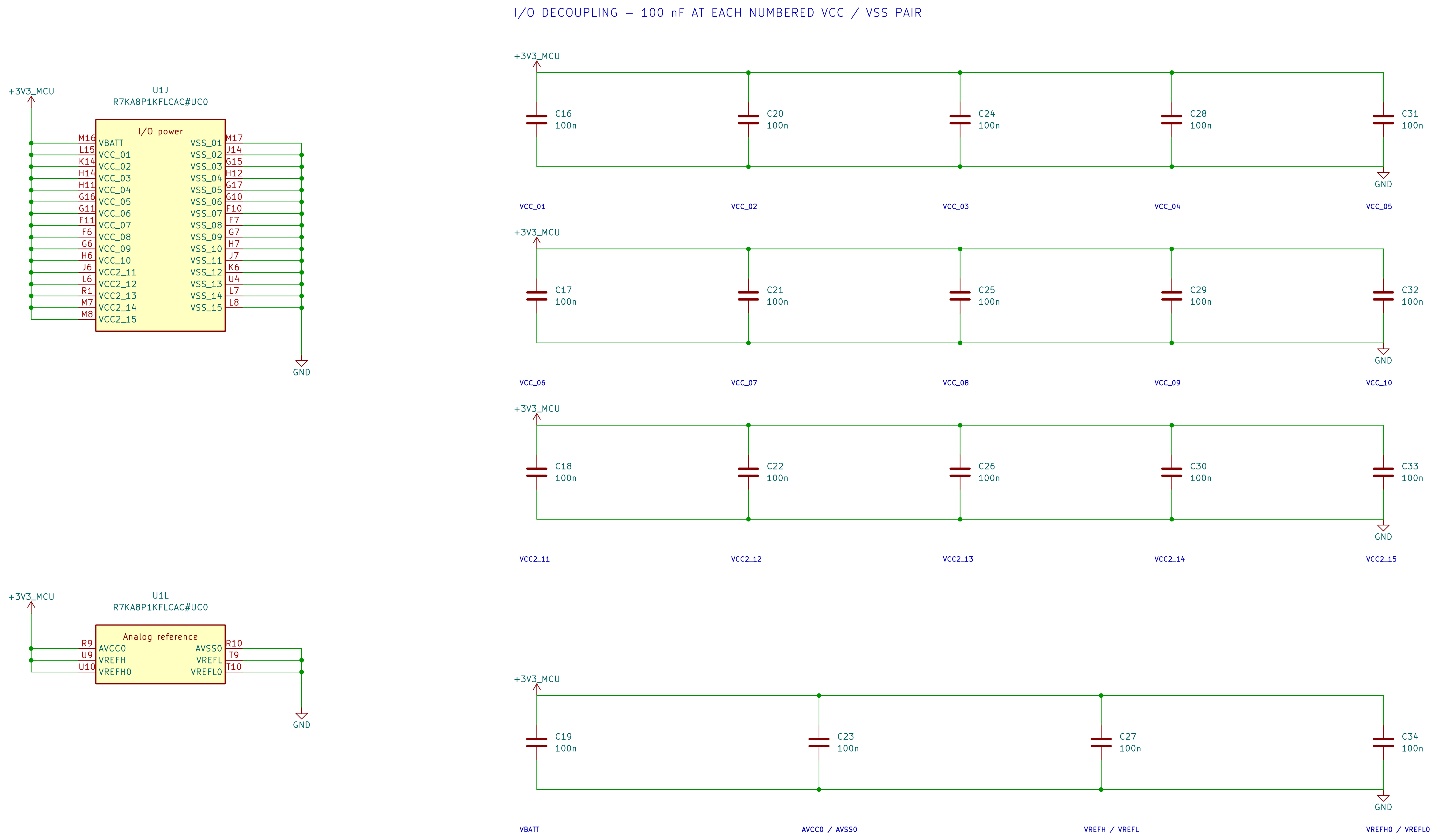

VCC and VCC2 use 3.3 V. All returns use one board ground plane.
VBATT follows the main rail; no separate backup–cell supply.
Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs.
USB/MIPI supplies remain on the interface sheet and are not yet wired.
HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862.
This is an electrical draft; capacitor MPN qualification remains open.

RA8P1 e-reader
Sheet: /RA8P1 IO and analog supplies/
File: mcu_supply_power.kicad_sch

Title: RA8P1 I/O and analog supplies

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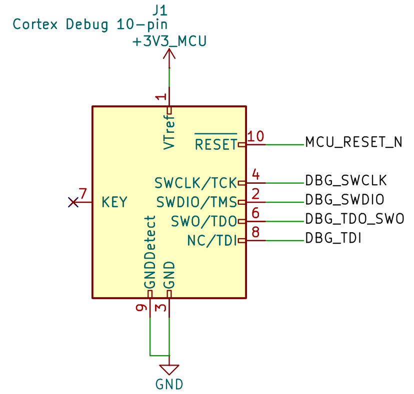

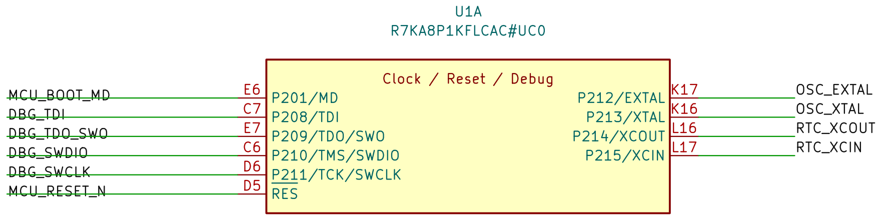

CLK-001: Y1 24 MHz / Murata XRCGB24M000F3M19R0
Crystal CL = 6 pF; effective Cstray = 2 pF ASSUMED.
CL = C35*C36/(C35+C36) + Cstray
C35 = C36 = 2*(6-2) = 8 pF (NP0, +/-0.5 pF).
Check: 8*8/(8+8) + 2 = 6 pF. At fixed stray: 5.75..6.25 pF.
Different crystal or changed parasitics: recalculate and rematch.
Initial matching only: verify frequency, startup and drive <= 300 uW.
Full derivation + Python check: ../design/clock_calculations.md

CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pF; Cstray=4 pF ASSUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38=2*(6-4)=4 pF, +/-0.25 pF NP0.
Check: 4*4/(4+4)+4=6 pF.
Fixed-stray tolerance: 5.875..6.125 pF.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md

NO capacitor on RES: it must track VCC for internal POR.
External reset drivers must be open-drain/open-collector.
Hold RESET low >= 3 ms, then release after VCC is valid.
Hold BOOT during external reset release for SCI/USBFS mode.
Normal MD high; SWD/JTAG boot requests occur during RES low.
Device lifecycle / authentication restrictions still apply.

References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3-4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.

Sheet: /RA8P1 clocks, reset and debug/
File: mcu_clocks_debug.kicad_sch

Title: RA8P1 clocks, reset and debug

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